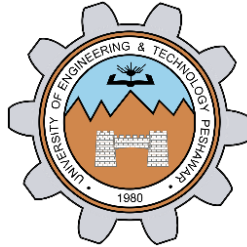


Design of a Light switch

LAB # 09



Spring 2024

CSE-308L Digital System Design Lab

Submitted by: **Ali Asghar**

Registration No.: **21PWCSE2059**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Asif Ali Khan

Date:

23rd May 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Objectives:

This lab will enable students to:

- Build a light switch controller such that when the push button is pressed the light if “off” turns “on” and if “on” turns “off”.

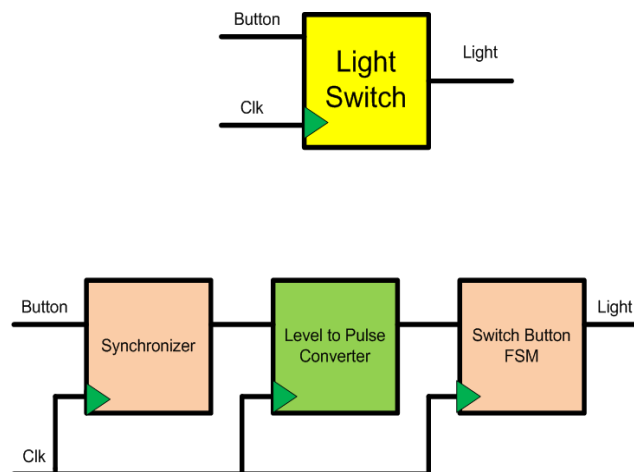
Software used:

- Xilinx ISE

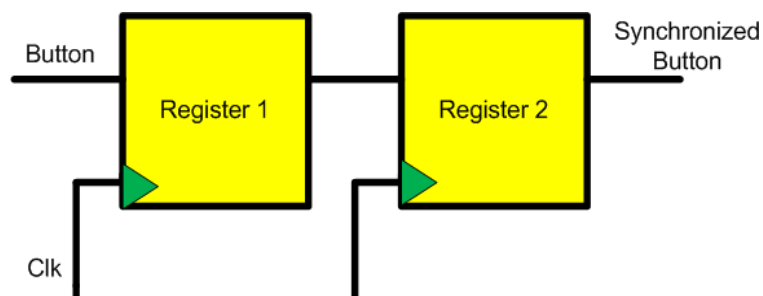
Block Diagram:

The top level block diagram is shown in the following figure. The second diagram is a more detailed view of the top level diagram. Synchronizer circuit is used to avoid the metastability problem which arises when synchronous digital system are fed with an asynchronous input. The level to pulse converter is there to convert level input from a push button into a pulse that is high for only one clock cycle. The switch button FSM is actually implementing the state machine for the requirement i-e when the push button is pressed the light if “off” turns “on” and if “on” turns “off”.

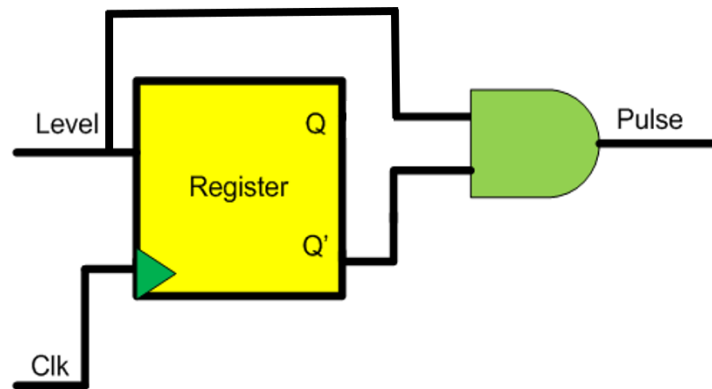
I/O connection: Connect the button input to a push button on the S3BOARD and light output to LED.



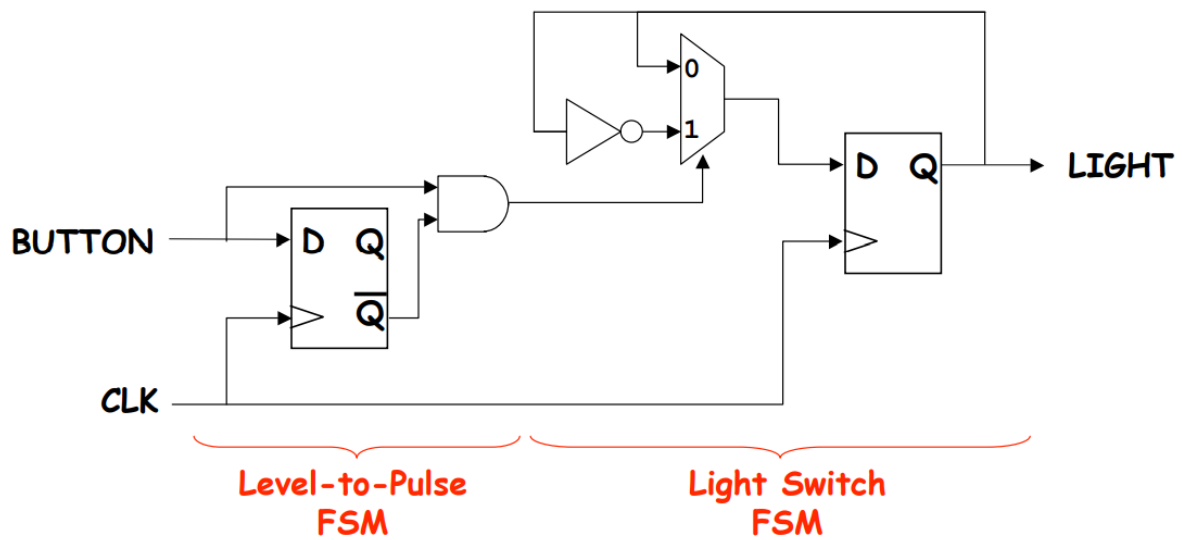
Synchronizer Circuit:



Level to Pulse FSM:

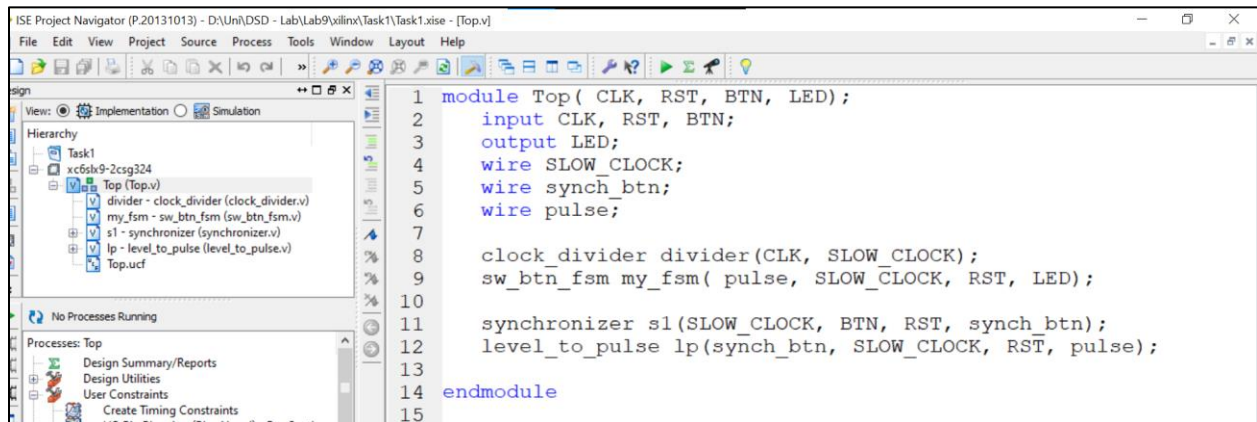


Switch Button FSM:



Lab Task:

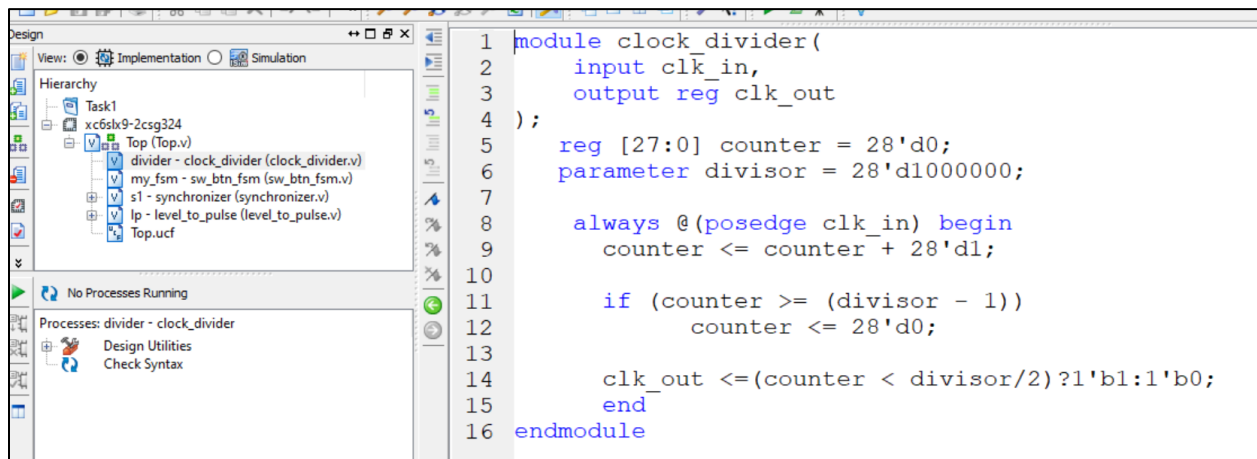
Code:



The screenshot shows the ISE Project Navigator window with the 'Top' module selected in the hierarchy. The code for the 'Top' module is displayed in the main editor area. The code defines a module 'Top' with inputs CLK, RST, and BTN, and output LED. It includes wires for SLOW_CLOCK, synch_btn, and pulse. The module instantiates three sub-modules: clock_divider, my_fsm, and level_to_pulse. The code is as follows:

```
1 module Top( CLK, RST, BTN, LED);
2   input CLK, RST, BTN;
3   output LED;
4   wire SLOW_CLOCK;
5   wire synch_btn;
6   wire pulse;
7
8   clock_divider divider(CLK, SLOW_CLOCK);
9   sw_btn_fsm my_fsm( pulse, SLOW_CLOCK, RST, LED);
10
11   synchronizer s1(SLOW_CLOCK, BTN, RST, synch_btn);
12   level_to_pulse lp(synch_btn, SLOW_CLOCK, RST, pulse);
13
14 endmodule
15
```

Figure 1.1 top module



The screenshot shows the ISE Project Navigator window with the 'clock_divider' module selected in the hierarchy. The code for the 'clock_divider' module is displayed in the main editor area. The code defines a module 'clock_divider' with input clk_in and output reg clk_out. It includes a 28-bit counter and a divisor parameter. The module uses an always block to increment the counter on the rising edge of clk_in. The code is as follows:

```
1 module clock_divider(
2   input clk_in,
3   output reg clk_out
4 );
5   reg [27:0] counter = 28'd0;
6   parameter divisor = 28'd1000000;
7
8   always @(posedge clk_in) begin
9     counter <= counter + 28'd1;
10
11     if (counter >= (divisor - 1))
12       counter <= 28'd0;
13
14     clk_out <= (counter < divisor/2) ? 1'b1 : 1'b0;
15   end
16 endmodule
```

Figure 1.2: Clock DividerCode

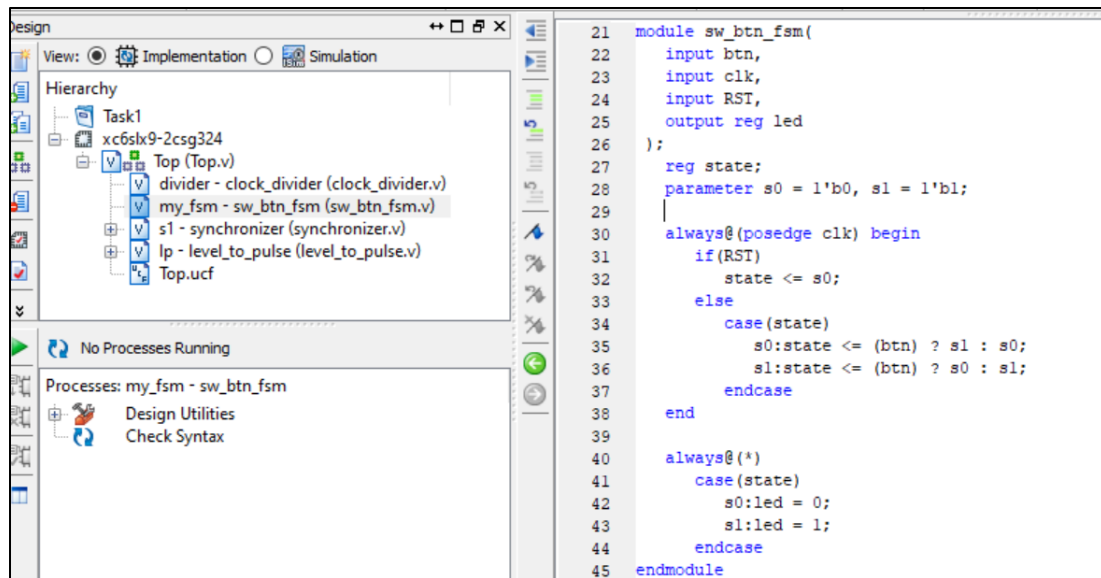


Figure 1.3: Switch Button FSM Code

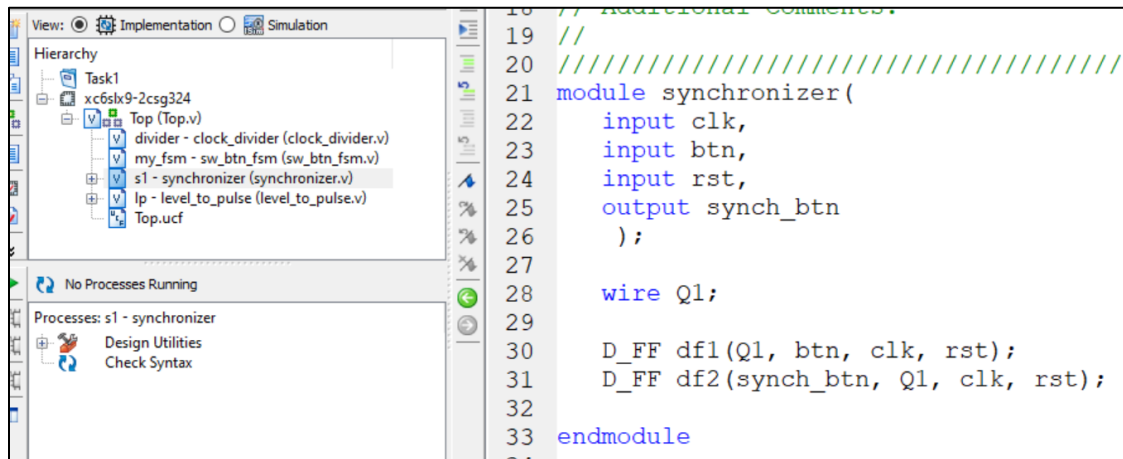


Figure 1.4: Synchronizer Code

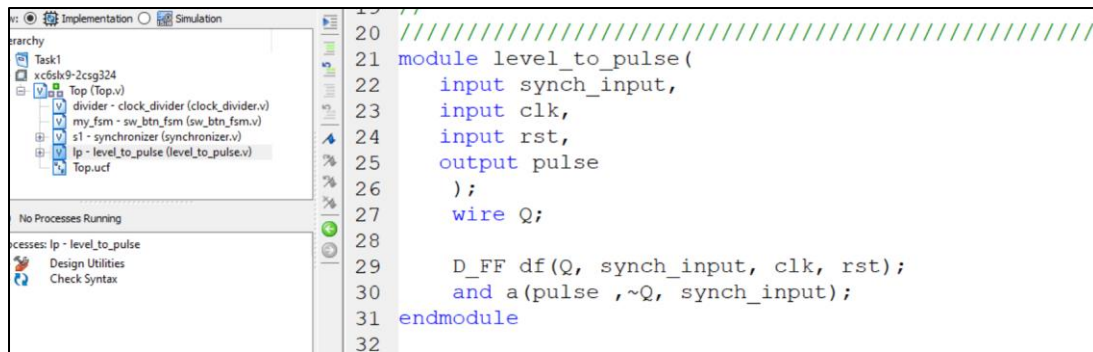


Figure 1.5: Level to Pulse Converter Code

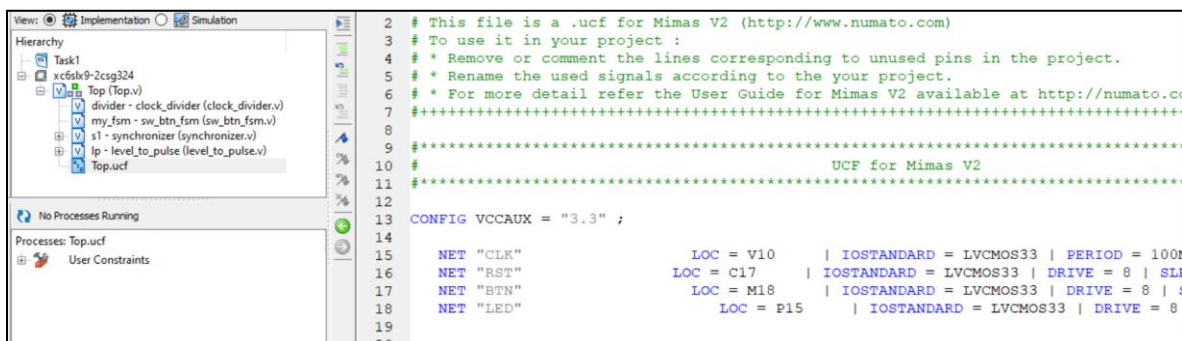


Figure 1.6: UCF File

Output:



Figure 2.1: Light Switch Implementation on Spartan6 FPGA Board (Off State)

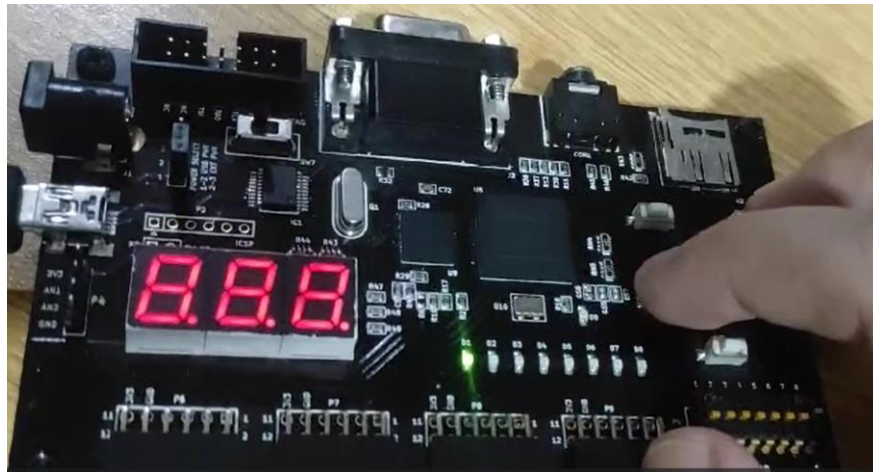


Figure 2.2: Light Switch Implementation on Spartan6 FPGA Board (On State)